



Optimal Sizing and Analysis of a Hybrid Solar-Wind-Diesel-Battery Energy System Considering Distinct Tracking Systems in Iraq

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Received: 03-18-2026

Revised: 04-28-2026

Accepted: 05-09-2026

Citation: A. M. S. Al-Bayati and H. H. Abdulkhudhur, "Optimal sizing and analysis of a hybrid solar-wind-diesel-battery energy system considering distinct tracking systems in Iraq," *Int. J. Energy Prod. Manag.*, vol. 11, no. 2, pp. 323–345, 2026. <https://doi.org/10.56578/ijepm110204>.



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Abstract: The power shortage and dependence on fossil fuels are the two key challenges facing the electricity sector in Iraq. Employing renewable energy sources may provide a promising solution for the electric power industry issues across the country. This necessitates extensive research efforts from multiple perspectives to establish their region-specific feasibility and viability. This study introduces an optimal sizing of a hybrid renewable energy system (HRES) under multiple solar tracking strategies for a residential community in Iraq. The proposed HRES integrates solar photovoltaic (PV), wind turbines, diesel generator, battery bank, and power converter. The solar tracking configurations employed in this work are horizontal-axis continuous adjustment (HACA) tracker, vertical-axis continuous adjustment (VACA) tracker, and two-axis (TA) tracker. The focus is on quantifying the viability of the solar tracking systems and their impacts on the overall system optimality and performance, considering the load demand, renewable resources, and cost data in the region. The results show that the optimal design of the HRES with the VACA solar tracking provides the best economic performance, yielding the lowest total net present cost and cost of energy, \$2.99 million and 0.134 \$/kWh respectively, the highest return on investment and internal rate of return values, 15.5% and 19.8%, respectively, and the shortest simple payback period at 4.7 years. Furthermore, the results revealed that the HACA solar tracking and the TA solar tracking do not lead to a lower overall system cost compared to the no solar tracking configuration. From the technical standpoint, the results demonstrate that an enhancement in solar capturing results in a reduction in solar PV array size within the hybrid system and simultaneously increases its efficiency in electricity generation. Environmentally, it was found that advanced solar tracking systems may not necessarily provide meaningful environmental benefits. The findings and the conducted analyses presented in this paper highlight the significance of assessing solar tracking technologies at the hybrid system level to carefully determine their viability from economic, technical, and environmental perspectives.

Keywords: Hybrid renewable energy systems; Solar photovoltaic; Wind turbine; Diesel generator; Battery storage; Hybrid Optimization of Multiple Energy Resources; Net present cost; Solar tracking systems

1 Introduction

The global demand for energy has been steadily rising in recent years. This is primarily because of the higher demand for energy services needed by continually growing and well-off population. Based on the latest report from the International Energy Agency, the global demand of energy has gone higher by about 60% during the period from 2000 to 2024, as shown in Figure 1. Despite extensive energy transition and transformation efforts, fossil fuels still represent the predominant energy sources, accounting for approximately four-fifths of total global energy demand in 2024 [1]. Coal, oil, and natural gas exhibit major disadvantages such as greenhouse gas emissions, environmental pollution, resource depletion, and economic volatility. Therefore, renewable energy stands as a perfect replacement to mitigate the drawbacks associated with fossil fuels. Solar and wind energy, as renewable sources, possess strong potential in electric power generation, particularly when integrated with traditional fossil fuel power plants. In the year 2024, renewable power installations grew globally by 22% to reach approximately 685 GW. Solar energy accounted for the largest share of this growth in global renewable power installations [2]. Here, it should be noted that

the bulk of global renewable power capacity is situated in China, European Union, United States, India, and Brazil. Therefore, the active involvement of other countries, especially developing ones, in renewable energy transition is necessary toward clean and sustainable electric power generation.

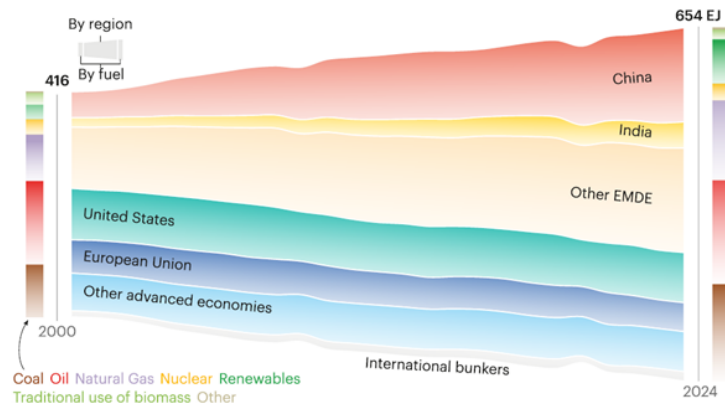


Figure 1. Global energy demand for the period from 2000 to 2024. Reproduced from Ref. [1] under the Creative Commons CC BY 4.0 license

Over the past 35 years up to now, the electricity shortage has represented a critical issue in Iraq. It affects a wide range of sectors within the country, such as industrial production, business developments, financial corporations, and the daily lives of people. Lack of diversification of electric power generation sources, along with old power infrastructures as well as the persistent high demand for electricity, led to severe deterioration in power generation, transmission, and distribution across the country. Electricity generation in Iraq still depends substantially on oil and natural gas [3]. This situation is unfortunate because of the national disinterest in the renewable energy technologies over the past years. Despite the fact that the country is rich in renewable energy resources, solar photovoltaic (PV) and wind turbine (WT) technologies remain largely absent from the national electric power grid. Thus, it is necessary for Iraq to initiate rules and guidelines that facilitate and encourage the investments in renewable energy technologies. In this regard, distributed hybrid small-scale power generation systems, integrating PV array, WTs, diesel generators (DGs), and batteries, may represent a powerful solution to deliver the electricity and meet the demand locally.

Although the renewable energy sources promise clean power and energy security, their intermittent nature and fluctuations pose a significant challenge. Solar PV modules and WTs produce only when the solar irradiation and wind are available, respectively. Hybrid renewable energy systems (HRESs) are developed as an approach to overcome that obstacle through integrating multiple complementary resources. The combination of solar PV with WTs and DGs, together with battery storage, leads to improvement in the reliability and stability of electricity generation compared to individual renewable sources [4–6]. Energy storage systems have a significant role in the transition from the intermittent to the reliable condition of renewables-based systems. These systems are responsible for storing the excess generation in the periods when the solar and wind are available and releasing it when resources fluctuate or become unavailable [7, 8]. With respect to enabling electricity access to remote regions, HRESs provide an effective solution to supply electric power for such areas. The remote and rural areas are normally far away from traditional power plants, which makes central grid extension expensive and difficult. Techno-economic evaluations of HRESs show that proper combinations of different energy sources can bring many advantages such as lowering cost of energy (COE), reducing dependence on fossil fuels, decreasing environmental emissions, and ensuring consistent electricity supply to the rural communities [9–14].

The appropriate sizing of HRES components is crucial from functional, economic, and sustainability perspectives. It determines the capability of a system to reliably supply the load with the lowest COE while achieving investment goals. Improper sizing can result in multiple issues, including high installation and operation costs, increased cost of electricity, and reduced environmental reliability [15, 16]. For these reasons, researchers have developed and used many modeling and optimization approaches [17–21]. Techniques such as Particle Swarm Optimization (PSO), Genetic Algorithms, and Grey Wolf Optimization have been used in HRES optimizations because of their ability to search large solution spaces and determine the optimal combination of energy sources [22]. In Ref. [23], the Grey Wolf Optimization algorithm was combined with Cuckoo Search algorithm for the purpose of optimal sizing of a hybrid system composed of PV array, WT, DG, and batteries. A mixed-integer nonlinear programming was used in Ref. [24] to formulate two optimization tasks, one for sizing the HRES and the other for ensuring an effective operation of the system. In another work [25], a multi-objective optimization strategy was employed to optimize both sizing and energy management of a HRES that combines solar PV, WTs, and multiple energy storage technologies. In the context of standalone HRESs, researchers follow methodological diversity with a concentration on providing

sustainable electricity to remote and rural regions. In Ref. [26], the authors used PSO to obtain an optimal mix of solar PV, WTs, biomass, battery storage, electrolyzers, and fuel cells in Kern County in the United States. Another work applied Bonobo Optimizer to design an off-grid HRES feeding a remote region in northern Saudi Arabia [27]. A techno-economic analysis was performed in Ref. [28] using Hybrid Optimization of Multiple Energy Resources (HOMER) with the objective to minimize the NPC and COE for a HRES incorporating multiple types of energy storage systems. In a different recent work [29], techno-economic and sensitivity analyses were also conducted using HOMER to determine the most favorable standalone HRES for a university in Bangladesh. In Ref. [30], the authors optimized a hybrid system comprising solar PV, WTs, and batteries using HOMER to power a large-scale seawater desalination in Saudi Arabia. The study demonstrated the potential of HRESs to reduce relying on traditional fossil fuels based desalination and mitigate associated environmental pollution. In grid-connected HRESs, the direction is a little different. As they connect with large electrical grids, optimization of these systems has to take into account not only the internal balance of the power sources and storage elements, but also the grid constraints, economics, and dynamics [31–33]. A recent study compared a grid-connected HRES with a stand-alone system from both reliability and cost perspectives. The study revealed that minimizing the system cost alone may not result in the most efficient system design and highlighted the significance of achieving balance between capacity sufficiency, cost, and reliability [34].

When it comes to solar PV module performance, the ability to capture solar irradiance to the greatest extent is of paramount importance. The generated power from a PV module relies significantly on the amount of sun's rays that reach the module surface directly. A fixed-tilt PV module is directed in only one orientation, normally positioned with an angle that reflects the annual average climate conditions at the location. However, it will not be able to track the continuous changes in the sun's position. Conversely, solar trackers can modify the orientation of the PV modules in order to track the sun during daylight hours. Because the sunlight angle and intensity changes during the day as well as the daylight duration and sunlight intensity variations across the seasons, solar tracking systems lead to produce more power throughout the year. Two main configurations of solar trackers were reported in existing studies, namely single-axis and dual-axis tracking systems. A single-axis solar tracker changes either the azimuth angle to make the PV module track the sun's east to west movement all day long, or the tilt angle to make it follow the sun's elevation. A dual-axis solar tracker, in turn, can change both the angles, allowing the PV module to face the sun's exact position [35, 36]. The performance advantages of solar tracking systems have been reported in prior publications in comparison with fixed systems [37–41]. The authors in Ref. [42] conducted an Internet of Things-based experimental study to quantify the improvement in power production resulting from the use of dual-axis solar tracking in the coastal region of Ecuador. The results showed that the PV panels with dual-axis tracking yielded 19.62% more energy than the fixed systems. In Ref. [43], technical and economic analyses revealed that a PV system with single-axis tracking in Cyprus can produce about 20% to 30% more energy in comparison with a PV system with fixed setup. In addition, the results presented in Ref. [44] highlighted that a single-axis solar tracker enables the PV system to produce roughly 20% more electricity compared to a fixed system. These studies confirm that the solar tracking systems can deliver higher energy production contrasted with fixed tilt configurations. However, in the context of HRESs optimization and management, the tracking systems may not always offer an economical solution compared to the fixed-tilt system [45].

Although considerable research has been done globally on the effect of solar tracking systems on PV system's energy generation improvement, a thorough review of the literature shows significant research gaps. First, many of the existing works focus on solar-only systems. These works present energy enhancement for solar PV modules without comprehensively determining the impacts of solar trackers on component sizing and economic performance at the system level. As solar PV technology is often employed with other energy sources and storage systems, conducting broader system-level studies on solar tracking systems is significantly important. Second, generalization of research findings on solar tracking systems to other locations is not feasible. This is because the performance of solar trackers may vary based on geographical location, climate conditions, and local economics. Third, despite its importance, system-level performance assessment of solar tracking systems in Iraq remains unexplored. Iraq is rich in solar resource with high direct irradiance and long daylight hours, particularly during the summer and fall seasons. Therefore, extensive research efforts are needed to quantify how solar trackers would perform as part of a HRES and to what extent they may shift the sizing and energy cost optimization of hybrid systems in this region.

In this paper, an optimized design and sizing of a HRES that combines solar PV, WTs, DG, battery bank, and power converter for a residential community in Dakuk region, Iraq is presented. Different solar tracking systems are studied at the HRES level and their impacts on the system optimality and performance are quantified and reported. Motivated by the aforementioned research gaps, this work makes several significant contributions to the field of HRESs in Iraq. First, it introduces a comprehensive system-level assessment of multiple solar tracking systems involving horizontal-axis continuous adjustment (HACA) tracker, vertical-axis continuous adjustment (VACA) tracker, and two-axis (TA) tracker, under Iraq unique geographical, climate, and economic conditions. Second, of particular importance, this work integrates the solar tracking systems into HRES optimization process and quantifies their impacts on component

sizing, system economics, and environmental emissions, which many papers overlook. Third, this work contributes to resolving the continual power shortage problem in Iraq by showing how the optimized design of HRESs can supply the load demand of local residential communities. Finally, the study helps to accelerate the deployment of HRESs in the country through clarifying trade-offs between system complexity, technical performance, and long-term economic viability. The remainder of this paper is structured as follows. Section 2 introduces the methodology and describes the proposed HRES in this work. In Section 3, the findings of detailed economic, technical, environmental, and sensitivity analyses under various solar tracking strategies are reported and discussed. Finally, the key findings are provided in Section 4.

2 Methodology

This section presents a systematic methodology followed to design and optimize a HRES integrated with different solar tracking systems, with the aim of quantifying the viability and impacts of the solar trackers on the overall system optimality and performance in Iraq. Rather than utilizing complex and computationally intensive algorithms, HOMER software was employed in this work. It is a robustly developed, widely recognized, and commonly adopted in academic research and real-world applications. HOMER is engineered to handle complex hybrid configurations, a wide range of technologies, and long-term technical, financial, and environmental assessments [46–49]. To model the system, the site location and its specific renewable resources are first defined. Then, the load profile that reflects the actual demand for a residential community is prepared. Next, the detailed technical and economic specifications for each component of the HRES, including solar PV modules, WTs, DG, batteries, and power converter, are collected and introduced. The system is set up and simulated under several solar tracking configurations, including HACA, VACA, and TA trackers, and additionally without solar tracking.

The main economic performance metrics that this work relies on are the net present cost (NPC) and COE. They assist in arranging the feasible configurations of the system and in offering valuable economic insights toward achieving a least-cost solution. The total NPC is a central indicator that represents the present value of the system's installation, operation, and maintenance costs across the project's lifespan minus the present value of the system's lifespan revenues. It can be calculated by:

$$NPC = \frac{C_{ann,tot}}{CRF(i, n)} \quad (1)$$

where, $C_{ann,tot}$ is the total annualized cost, i is the real discount rate, n is the lifetime of the project, and CRF is the capital recovery factor which can be determined by:

$$CRF(i, n) = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (2)$$

The COE represents the average electricity cost per kWh and is calculated by:

$$COE = \frac{C_{ann,tot}}{E_{served}} \quad (3)$$

where, E_{served} is the total yearly electrical demand met [50, 51]. In addition to the NPC and COE, there are other important economic metrics, namely return on investment (ROI) and internal rate of return (IRR), that represent profitability measures for a HRES. The ROI represents the annual average return from the proposed system relative to a base system. The ROI is computed by dividing the average of annual difference in cash flows between the proposed system and a base system, by the difference between the capital costs of the same systems, and it is given by:

$$ROI = \frac{\sum_{i=0}^n (C_{i, base} - C_i)}{n(C_{cap} - C_{cap, base})} \quad (4)$$

where, $C_{i, base}$ is the nominal annual cash flow of the base system, C_i is the nominal annual cash flow of the proposed system, C_{cap} is the capital cost of the proposed system, and $C_{cap, base}$ is the capital cost of the base system. The IRR is defined as the discount rate at which the NPC of the base case and current system is equal. It is computed by finding the discount rate that makes the present value of the difference of the system's two cash flows equal zero [50, 51]. These metrics together help the investors and decision makers to efficiently elect the most cost-effective and profitable system configuration.

2.1 Geographical Setting, Renewable Potential, and Load Profile

The angle of incidence of sun irradiance on the PV modules significantly influences their performance. In light of this, the essential objective of this paper is to study and quantify the effects of solar tracking systems on the

optimal design and performance of HRESs in Iraq. Thus, Dakuk (Daquq) city is selected as the study location. The city is situated in a strategically important area at the northern Iraq within Kirkuk province. The exact position of the study lies at 35°12.0'N latitude and 44°28.0'E longitude, as depicted in Figure 2. With respect to the study location, the Dakuk region provides a suitable combination of renewable energy resources, particularly solar and wind, for designing and assessing HRESs under Iraq's weather conditions. The region records relatively high levels of solar irradiance and sky clarity across the year, along with dependable wind resources, as will be detailed later in this work. During the summer, the area exhibits hot and dry climate conditions along with long days, while during the winter, it experiences a milder weather with shorter days. The Dakuk region is also characterized by a semi-arid environment which is typical of many parts of the country. Furthermore, a comparison with other regions in Iraq provides further clarification of the relevance of the selected region. For instance, the southernmost city of Basra experiences very high levels of solar irradiance while the northernmost city of Duhok exhibits comparatively lower levels of solar irradiance with favorable wind speeds at certain times. The central and several southern and northern regions of Iraq, including Dakuk, generally exhibit balanced characteristics of adequate solar irradiance and suitable wind speeds. Therefore, all these characteristics render the Dakuk region to be a representative case for many Iraqi regions as well as a most suitable location to assess the impacts of different solar tracking strategies within HRESs in Iraq.



Figure 2. Geographical location of Dakuk city

The assessment of renewable resources at a site is an essential step prior to installing HRESs. A stable and complementary availability of solar irradiance and wind enables a HRES to operate in a resilient and efficient manner. To secure the quality of resource data, the measurements of meteorological information in this work were obtained from NASA database. The monthly average values of solar irradiance and clearness index is shown in Figure 3. The site exhibits an annual average of solar radiation of about 4.99 kWh/m²/day, with a steadily trend of increasing from January and peaking in June. The clearness index, which is a measure of atmospheric transparency and cloudiness, is the highest during the months from June to September, signifying clearer skies and stronger direct irradiance. The abundant solar irradiance and clear atmospheric in these months offer an optimal condition for the operation of PV modules and solar tracking systems. The ambient temperature is an another parameter that influences the performance of PV modules. Although the solar irradiance enables PV power generation, higher

values of temperature reduce the efficiency of the PV modules. The monthly average temperature extends from 7.46 °C to 35.55 °C, with an annual average of 21.50 °C, as illustrated in Figure 4. This temperature profile demonstrates a long period of hot weather during the summer, reaching its maximum in July, and relatively mild weather during the other seasons, with minimum values in January.

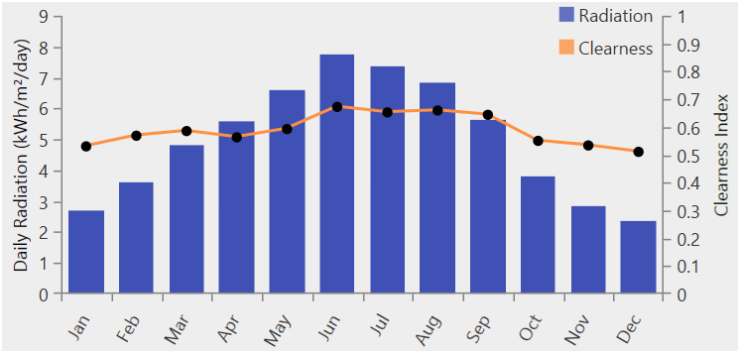


Figure 3. Monthly average values of solar irradiance and clearness index

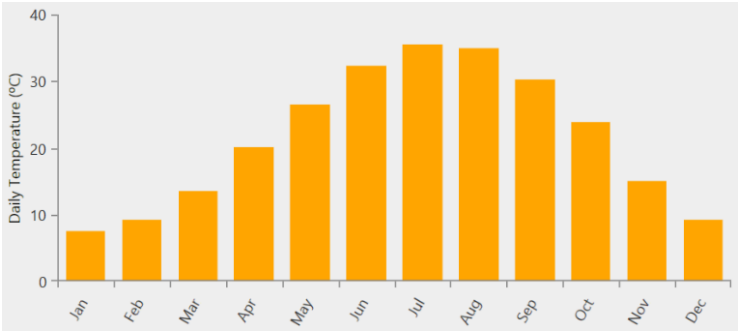


Figure 4. Monthly average values of air temperature

Alongside solar irradiance and temperature, dependable wind speed is a crucial parameter for a HRES installation. It frequently complements the presence of solar irradiance which allows the HRES to be designed as a reliable and efficient system. Figure 5 illustrates the wind speed profile for the location considered in this work. It demonstrates a steady monthly trend of the wind speeds throughout the year. The monthly average wind speed ranges from 4.60 m/s to 5.87 m/s, with an annual average of 5.12 m/s. Such trend of the wind data holds significant value because it reveals that the wind speed will act as a reliable and complementary contributor alongside the other resources within the hybrid system.

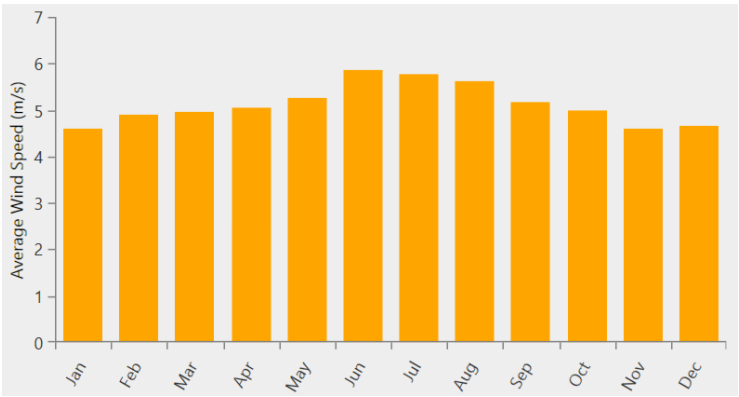


Figure 5. Monthly average values of wind speed

Broadly speaking, the load data of a particular location has a critical role in the optimization process. Careful estimation of the load profile is an important technical step to prevent the oversizing or undersizing of system components. Such improper sizing may lead to major issues such as high COE, power shortages, accelerated battery

degradation, and poor system reliability. In this paper, the load was methodically and accurately estimated for 30 houses in Dakuk city. It was built and prepared based on the types and quantities of appliances, daily usages, local consumption patterns in the region, and seasonal behavior. Moreover, 10% of random variability was introduced into the load profile in order to reflect the real-world demand fluctuations. The monthly average load profile is depicted in Figure 6. The load demand is relatively low during the winter and spring months, while it is the highest during the summer months due to air conditioning loads, and it goes down during the fall season. The daily average demand estimated at 4088.20 kWh/day with a peak of 533.64 kW.

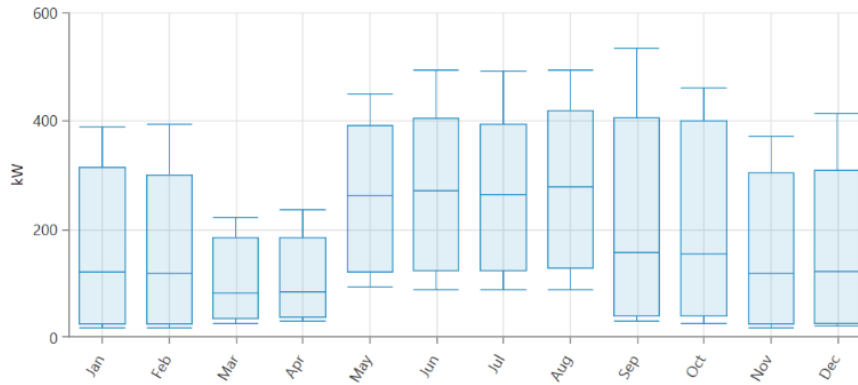


Figure 6. Monthly average load profile

2.2 Model Design and System Components

This subsection introduces the model of the proposed HRES along with the technical specifications and economic parameters of its components. Figure 7 illustrates a structural diagram of the proposed HRES. It comprises a solar PV installation with solar tracking system, WTs, DG, batteries, and converter. The system is designed to reliably supply electric power to a small residential community in the city of Dakuk. The solar PV array, WTs, and DG are responsible to generate the needed power to supply the demand. The batteries and a power converter are used to store the excess electricity and to perform power conversions, respectively. The HOMER software is employed as an optimization tool for optimally sizing the proposed system components under different solar tracking systems.

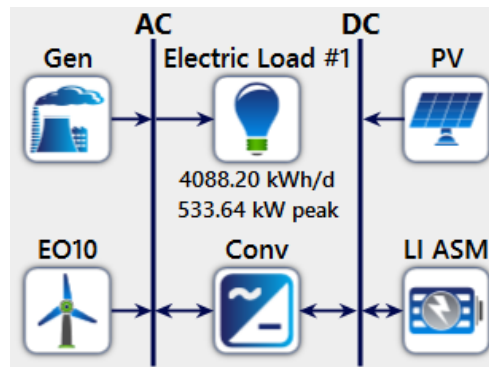


Figure 7. Schematic of the proposed hybrid renewable energy system

Note: AC = alternating current bus; DC = direct current bus; PV = photovoltaic; EO10 = wind turbine model type ; Conv = converter; LI ASM = lithium-ion battery.

2.2.1 Solar solar photovoltaic modules and tracking systems

In standalone PV systems, the number of PV modules is primarily determined based on the load demand and the levels of local solar radiation. However, within a hybrid setup, the PV capacity is sized in a complementary manner with the other energy sources as a result of economic and operational considerations. The power produced by a PV system is mainly affected by the amount of solar radiation, the PV modules' operating temperature, and the orientation and tilt of the PV modules. The solar PV output power (PPV) can be determined by the following

equation [52]:

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha_P (T_C - T_{C,STC})] \quad (5)$$

where, Y_{PV} , f_{PV} , G_T , $G_{T,STC}$, α_P , T_C , and $T_{C,STC}$ is the rated capacity of the PV array, the PV derating factor, the instantaneous value of the solar radiation, the solar radiation at standard test conditions, the temperature coefficient of power, the instantaneous value of PV cell temperature, and the PV cell temperature under standard test conditions. The capital and replacement costs of the solar PV array in this paper were selected to be \$750 for a 1 kW capacity. Also, the yearly cost of the operation and maintenance was assumed as \$10 for every 1 kW of PV capacity. These cost values are grounded on real market data recorded in Iraq, which provides rigor to the conducted economic analysis. The PV array in this work is sized using the HOMER Optimizer.

A solar tracker is a mechanism comprising mechanical and control setup that enables the PV module to follow the sun position in the sky in order to keep the module perpendicular to incoming solar radiation. The majority of PV systems today is installed with fixed-position PV modules due to their simple configuration and low operation and maintenance requirements. In fixed position PV systems, the PV modules are installed at a constant tilt angle and azimuth. The constant value of the tilt angle in such systems is commonly selected based on the latitude of the location or an annual optimum angle. However, the main drawback of fixed installation is that a considerable amount of available solar irradiance could lose because of inefficient angle of incidence. Solar tracking systems, on the other hand, enhance solar irradiance capture, but they also involve higher costs. This tradeoff demands thorough exploration to determine the economic viability of solar tracking systems within HRESs and how they can impact the overall system optimization. In this work, the proposed HRES is studied under different solar tracking techniques, specifically fixed tilt (no solar tracking), HACA tracking, VACA tracking, and TA tracking. In HACA tracking system, the PV module rotates around a horizontal axis where the tilt angle changes continuously while the azimuth angle stays fixed. Conversely, in VACA tracking system, the PV module rotates around a vertical axis where the azimuth angle varies continuously while the tilt angle remains constant. In TA tracking system, the PV module turns around both horizontal and vertical axes where both the tilt and azimuth angles change continuously throughout the day. The capital and replacement costs of both the HACA and VACA tracking systems were considered to be 200 \$/kW along with an annual operation and maintenance cost of 5 \$/kW. Further, the capital and replacement costs of the TA tracking system are 400 \$/kW and its annual operation and maintenance cost is 5 \$/kW. These costs are for the solar tracking systems exclusively, excluding the PV modules costs.

2.2.2 Wind turbine integration

Based on the wind speed data shown in Figure 5, this work considers the integration of wind power into the HRES. The persistent wind speeds at the location throughout the year make the WTs a good complement to solar PV array, particularly during nighttime or cloudy days. The power output from a WT is strongly correlated with the wind speed in a cubic relationship. It can be calculated using the following equation [53]:

$$P = \frac{1}{2} \rho C_p A V^3 \quad (6)$$

where, P , ρ , C_p , A , and V are the turbine power output, the air density, the turbine power coefficient, the rotor swept area, and the wind speed at the turbine hub height. Considering wind speed profile at the location, the WTs of the Eocycle EO10 Class III model type were selected in this research. The technical characteristics of this model include 10 kW of rated power, 16 m of hub height, 20 years of lifetime, 2.75 m/s of cut-in wind speed, and 20 m/s of cut-out wind speed. Moreover, both the capital and replacement costs are \$35,000 per a turbine, in addition to a yearly operation and maintenance cost of \$350.

2.2.3 Diesel generator integration

The DG plays an essential role within the HRES. It serves as a supplementary power source within the system when the solar or wind resource is insufficient. Because of the dispatchable characteristics of the DG, it can increase the reliability and stability of the system by supplying the load demand during periods of renewable intermittency or sudden load spikes. However, its main drawback is that it depends on fossil fuel and thereby introduces greenhouse gas emissions. In this work, a DG is integrated to the system to secure the reliability and resilience. The considered DG has a 30% of minimum load ratio and 20,000 h of lifetime. Its initial capital and replacement costs are 275 \$/kW with a 0.02 \$/kW of operation and maintenance cost for every hour of operation. These cost data were guided by the Iraqi market for DGs. The price of the fuel was set at \$0.308/L as determined by the official fuel pricing in the country [54].

2.2.4 Battery storage system

Energy storage systems are the backbone of HRESs through their function in absorbing excess electricity or instantly delivering power to the load. They reduce the fluctuations between the generation and consumption within the system and secure a continual power supply when renewable availability is low. A battery bank is integrated in this work into the system to interact continuously with the solar PV array, WTs, and DG to ensure the reliability and efficiency of the system. The real-world performance of batteries is significantly affected by many influences like temperature, depth of discharge, charging rates, aging mechanisms, and others. Therefore, it is important to properly size and manage the battery bank in a hybrid system to ensure the reliability and long-term sustainability. In this work, a battery of the lithium-ion type was selected with technical characteristics including a 3.7 V nominal voltage, a 1.02 kWh nominal capacity, and a 276 Ah maximum capacity. Further, its maximum and minimum operating temperatures are 60 °C and 0 °C, respectively. In addition, both the capital and replacement costs are \$350 per a battery with a yearly operation and maintenance cost of \$10. Here, it is worth mentioning that these considered values are grounded on real data from the Iraqi battery market.

2.2.5 Power converter integration

In the HRESs, the power converter serves as an interface between the power sources, battery units, and the load. It basically manages the flow of energy between the direct current (DC) and alternating current (AC) busses within the system. In most of HRES architectures, the power converter operates in a bidirectional mode, converting DC power into AC power and vice versa as needed. Selecting the appropriate power converter type for a HRES depends mainly on the system architecture, the type of energy sources within the system, the load profile, and the efficiency requirements. In this work, the power converter is optimally sized by HOMER based on the power flow between the system components depicted in Figure 7. From a technical perspective, the selected power converter in this work has a life cycle of 15 years and an efficiency of 95%. Economically, its capital and replacement costs were considered to be 300 \$/kW with a yearly operation and maintenance cost of 3 \$/kW.

3 Results and Discussion

This section presents the optimization and sensitivity analysis results of the proposed HRES under different solar tracking configurations. The aim is to quantify the effect of solar trackers on the system's optimality and performance as well as determining the most efficient solar tracking system for use in Iraq. Toward this end, the proposed system is studied and compared under various tracking mechanisms. These include the system with no solar tracking that is considered the baseline system, the system with the HACA solar tracking, the system with the VACA solar tracking, and the system with the TA solar tracking. The system in each case is modeled under identical load profile, resource conditions, technical data, and economic parameters. A comprehensive system-level analysis in each case is conducted from economic, technical, and environmental perspectives. Moreover, a sensitivity analysis of both the fuel price and nominal discount rate is performed to further assess the economic viability of the tracking systems. The obtained results and performed analyses are reported and discussed in the next subsections. Here, it is important to highlight that, in some parts of this work, the system with no solar tracking is referred to as Optimal System 1, while the systems with the HACA, VACA, and TA solar trackers are referred to as Optimal System 2, Optimal System 3, and Optimal System 4, respectively.

3.1 System-Level Economic Analysis of Solar Trackers

The key economic metrics of the HRES with no solar tracking are provided in Table 1. The system has a total NPC of \$3.02 million and a COE of 0.135 \$/kWh. In addition, it has a yearly operating cost of \$110,896. These economic parameters are an indication of the financial viability of the system. The NPC breakdown of the system with no solar tracking is presented in Table 2. The WTs have the highest capital investment with an initial cost of \$665,000, which reflects a substantial upfront expenditure needed to exploit the wind energy with a considerable scale at the region. Also, among all the components, the WTs show the highest replacement cost about \$279,197, which is an indication of a long-term financial requirement. After the WTs, the solar PV array has the second highest capital cost of \$213,263. However, the solar PV array has no replacement cost due to its long lifetime. Although the DG has the lowest capital cost compared to the WTs and solar PV array, it shows the highest operation and maintenance and fuel costs. This reveals the central role of the DG within the hybrid configuration in maintaining the system reliability. The battery bank has comparable capital and replacement costs, with capital cost at \$212,800 and replacement cost at \$205,109. Even though the power converter is relatively not expensive in comparison with the other components, it exhibits a steady contribution to the total NPC through the capital, replacement, and operation and maintenance costs. The economic results also reveal that there is some financial relief coming from the salvage costs of some of the components in the system. However, these salvage costs remain marginal compared to the other costs.

Table 1. Optimal system with no solar tracking, its configuration, and key parameters

Optimal System	Configuration and Key Parameters							Operating Cost (\$/yr)	Renewable Fraction (%)
	PV (kW)	WT	DG (kW)	Battery	Converter (kW)	NPC (\$M)	COE (\$/kWh)		
PV-WT-DG-Battery-Converter	284	19	590	608	360	3.02	0.135	110,896	61.4

Note: PV = photovoltaic; WT = wind turbine; DG = diesel generator; NPC = net present cost; COE = cost of energy.

Table 2. Breakdown of NPC for the optimal system with no solar tracking

System Components	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)
PV	213,263.40	0.00	42,448.31	0.00	0.00
WT	665,000.00	279,197.40	99,272.05	0.00	168,556.33
DG	162,250.00	155,052.02	297,344.42	705,518.56	48,801.92
Battery	212,800.00	205,109.74	90,763.02	0.00	62,144.25
Converter	107,973.75	56,316.48	16,118.46	0.00	12,163.52

Note: NPC = net present cost; PV = photovoltaic; WT = wind turbine; DG = diesel generator; O&M = operation and maintenance cost.

The economic performance of the HRES with HACA solar tracking are provided in Table 3. The total NPC of the system is \$3.05 million, the COE is 0.137 \$/kWh, and the yearly operating cost is \$112,293. They are slightly higher than the total NPC, COE, and yearly operating cost obtained for the system with no solar tracking. This means that the energy production from the PV modules that incorporate HACA solar trackers is slightly more expensive as the trackers' investment and maintenance costs outweigh the benefit in energy yield. Table 4 presents the NPC breakdown of the system with HACA solar tracking. It can be observed that the solar PV array exhibits a higher economic burden under horizontal tracking. The PV capital investment cost increased to \$220,938 compared to \$213,263 in the system with no solar tracking. In addition, the operation and maintenance cost also rose to \$52,076. These increased costs result from the HACA mechanism that adds more investment and ongoing maintenance costs to the solar PV array. The NPC breakdown reveals that the cost structure of the WTs stays the same between the two system configurations. This indicates that the dependence of the system on wind energy is not changed by introducing the HACA solar tracking. Moreover, the DG continues to have the highest long-term operational cost in both systems. In the case of HACA solar tracking, fuel cost increased slightly to \$710,106 compared to \$705,518 in the no solar tracking system. This reveals that even though the HACA solar tracking improves solar capture, the energy gains from solar PV array is not cost effective to reduce the DG operation. The costs of the battery bank and power converter show a marginal variation between the two cases. This system-level analysis highlights that the HACA solar tracking does not necessarily lower the total NPC and COE of the system unless it significantly increases the energy generation from the PV array and reduce the system's reliance on costly components.

Table 5 provides the key economic indicators of the HRES equipped with VACA solar tracking. The system has a total NPC of \$2.99 million, a COE of 0.134 \$/kWh, and a yearly operating cost of \$111,748. The system with VACA solar tracking has slightly lower total NPC and COE than the system with no solar tracking. However, it shows slightly higher yearly operating cost due to the additional operation and maintenance requirements associated with the VACA solar trackers. The NPC breakdown of the system with VACA solar tracking is provided in Table 6. Examining the NPC breakdown demonstrates that the capital cost of the solar PV array drops slightly to \$211,819 compared to \$213,263 in the no solar tracking system. This indicates that the VACA solar trackers benefit from the improved energy yield without leading to a significant increase in the investment related to the solar PV array. The results also show that there is a clear economic shift in the WTs costs. Compared to the system with no solar tracking, the capital and replacement costs of the WTs decrease to \$630,000 and \$264,502, respectively. This reduction indicates that the system with VACA solar tracking depends slightly less on wind energy. The fuel cost of the DG increased to \$714,028 as well as its operation and maintenance expenses rose to \$302,276. Despite this heavy reliance on the DG, the system still attains a net economic enhancement. The battery bank and power converter costs exhibit relatively minor variation compared to their costs in the no solar tracking case. This means that the VACA solar tracking system does not substantially interrupt the energy storage behavior or the power conversion requirements in the system. This analysis at the system level demonstrates the constructive role of the VACA solar tracking in reducing the total NPC and COE by helping to manage the costs more efficiently across the system components.

Table 3. Optimal system with HACA solar tracking, its configuration, and key parameters

Optimal System	Configuration and Key Parameters							Operating Cost (\$/yr)	Renewable Fraction (%)
	PV (kW)	WT	DG (kW)	Battery	Converter (kW)	NPC (\$M)	COE (\$/kWh)		
PV-WT-DG-Battery-Converter	233	19	590	616	359	3.05	0.137	112,293	61.1

Note: HACA = horizontal-axis continuous adjustment; PV = photovoltaic; WT = wind turbine; DG = diesel generator; NPC = net present cost; COE = cost of energy.

Table 4. Breakdown of NPC for the optimal system with HACA solar tracking

System Components	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)
PV	220,938.20	0.00	52,076.74	0.00	0.00
WT	665,000.00	279,197.40	99,272.05	0.00	168,556.33
DG	162,250.00	155,569.74	298,753.63	710,106.02	48,253.58
Battery	215,600.00	208,365.23	91,957.27	0.00	62,363.88
Converter	107,842.69	56,248.12	16,098.90	0.00	12,148.75

Note: NPC = net present cost; HACA = horizontal-axis continuous adjustment; PV = photovoltaic; WT = wind turbine; DG = diesel generator; O&M = operation and maintenance cost.

Table 5. Optimal system with VACA solar tracking, its configuration, and key parameters

Optimal System	Configuration and Key Parameters							Operating Cost (\$/yr)	Renewable Fraction (%)
	PV (kW)	WT	DG (kW)	Battery	Converter (kW)	NPC (\$M)	COE (\$/kWh)		
PV-WT-DG-Battery-Converter	223	18	590	608	364	2.99	0.134	111,748	60.9

Note: VACA = vertical-axis continuous adjustment; PV = photovoltaic; WT = wind turbine; DG = diesel generator; NPC = net present cost; COE = cost of energy.

Table 6. Breakdown of NPC for the optimal system with VACA solar tracking

System Components	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)
PV	211,819.82	0.00	49,927.47	0.00	0.00
WT	630,000.00	264,502.80	94,047.20	0.00	159,684.95
DG	162,250.00	156,851.20	302,276.67	714,028.03	46,882.74
Battery	212,800.00	204,390.22	90,763.02	0.00	62,913.11
Converter	109,062.27	56,884.22	16,280.95	0.00	12,286.14

Note: NPC = net present cost; VACA = vertical-axis continuous adjustment; PV = photovoltaic; WT = wind turbine; DG = diesel generator; O&M = operation and maintenance cost.

Table 7 presents the economic performance of the HRES equipped with TA solar tracking. The total NPC of the system is \$3.03 million, the COE is 0.136 \$/kWh, and the yearly operating cost is \$111,869. These values are slightly higher than those of the system with no solar tracking. This reveals that even though the TA solar tracking increases solar capture by the PV modules, its energy enhancement effects do not lead to economic gains at the system level. The NPC breakdown of the system with TA solar tracking is presented in Table 8. The capital cost of the solar PV array goes down to \$205,834 which is lower than that in the no solar tracking case. This means that the optimized system uses a smaller PV capacity to supply part of the load. This is also why the operation and maintenance cost is relatively lower than in the no solar tracking system. The costs of the WTs do not differ from those in the no solar tracking system. This indicates that the system with TA solar tracking cannot reduce its reliance on wind power. In addition, the results reveal that the DG still leads the operational economics as in the no solar tracking system. This is evident from the fuel cost, which increased to \$712,487, as well as from the operation and maintenance expenses,

which rose to \$300,162. This demonstrates that although the TA solar tracking maximizes solar capture, it does not contribute to reduce the operation of the DG within the system. The costs of the battery bank and power converter exhibit a marginal variation across both cases. In overall, this analysis illustrates that the complexity added by the TA solar tracking led to a decline in the economic returns at the system level.

Table 7. Optimal system with TA solar tracking, its configuration, and key parameters

Optimal System	Configuration and Key Parameters							Operating Cost (\$/yr)	Renewable Fraction (%)
	PV (kW)	WT	DG (kW)	Battery	Converter (kW)	NPC (\$M)	COE (\$/kWh)		
PV-WT-DG-Battery-Converter	179	19	590	616	357	3.03	0.136	111,869	61.0

Note: TA = two-axis; PV = photovoltaic; WT = wind turbine; DG = diesel generator; NPC = net present cost; COE = cost of energy.

Table 8. Breakdown of NPC for the optimal system with TA solar tracking

System Components	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)
PV	205,834.53	0.00	40,079.01	0.00	0.00
WT	665,000.00	279,197.40	99,272.05	0.00	168,556.33
DG	162,250.00	156,084.51	300,162.85	712,487.23	47,705.24
Battery	215,600.00	208,952.54	91,957.27	0.00	61,729.88
Converter	107,116.55	55,869.39	15,990.50	0.00	12,066.95

Note: NPC = net present cost; TA = two-axis; PV = photovoltaic; WT = wind turbine; DG = diesel generator; O&M = operation and maintenance cost.

A comparison of the NPC of the system components under different solar tracking strategies is depicted in Figure 8. It can be seen that the DG and WTs account for a large portion of the system total NPC in all the configurations. This reveals that the fuel consumption, operation and maintenance cost of the DG, and the wind power infrastructure stay the main contributors to the total NPC no matter the solar tracking strategy. A side by side comparison of the four system configurations demonstrates that the overall economic balance of the system varies depending on the tracking strategy considered. The system with the VACA solar tracking is the most cost-effective across the four configurations. This system achieved the lowest total NPC and COE despite the extra capital investment and operation and maintenance cost associated with the tracker. This reveals that the VACA solar tracking has the potential to enhance solar capture, thereby sufficiently increasing the energy yield from the solar PV array to compensate the additional investment and operating expenses. Therefore, from an economic standpoint, the VACA solar tracking represents the most advantageous option for the region compared with the other tracking strategies. In contrast, the HACA and TA solar tracking systems do not provide the same system-level economic advantage. The total NPC and COE of the system with either the HACA solar tracking or TA solar tracking are slightly above the no solar tracking case. This is because their additional capital investments and operation and maintenance requirements outweigh the energy benefits within the system.

Furthermore, a comparison of the investment performance indicators of the system under different solar tracking strategies is shown in Figure 9. The system with the VACA solar tracking has the highest ROI and IRR, at 15.5% and 19.8% respectively, as well as the shortest simple payback period at 4.7 years. This implies that the VACA solar tracking has the potential not only to reduce the long-term costs but also to accelerate the capital recovery of the system. The above analysis provides clear insights about the implications of the different tracking strategies on the system-level economic performance of the HRESs across the study region.

3.2 System-Level Technical Analysis of Solar Trackers

The technical specifications of the HRES with no solar tracking are provided in Table 1. The system configuration includes a solar PV capacity of 284 kW, 19 WTs, a 590 kW DG, 608 batteries, and a 360 kW power converter. The design of the system shows an effective management of the renewables' variability and system reliability. The renewable fraction of the system is 61.4% which reflects that a significant amount of the annual load demand is met by the renewable technologies. Therefore, the system is optimized such that to ensure the reliability on the one hand and to maintain a considerable percentage of renewable penetration on the other.

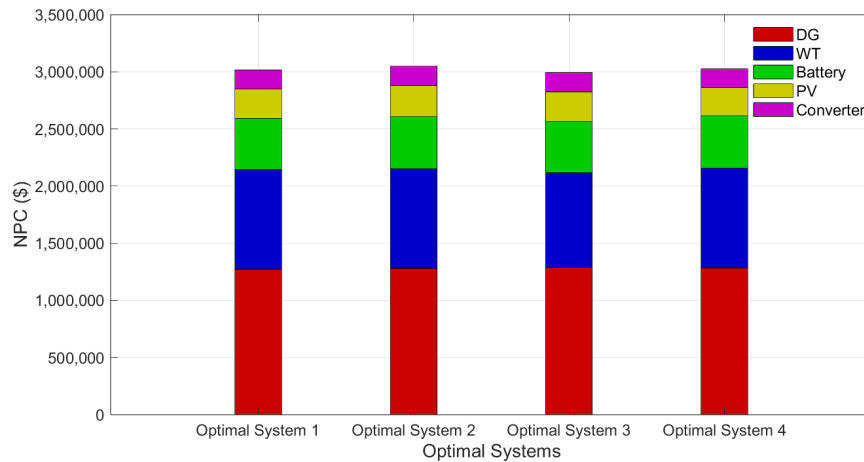


Figure 8. NPC comparison among components

Note: Optimal System 1, Optimal System 2, Optimal System 3, and Optimal System 4 represent the systems with no solar tracking, with horizontal-axis continuous adjustment solar tracking, with vertical-axis continuous adjustment solar tracking, and with two-axis solar tracking, respectively. PV = photovoltaic; WT = wind turbine; DG = diesel generator; NPC = net present cost.

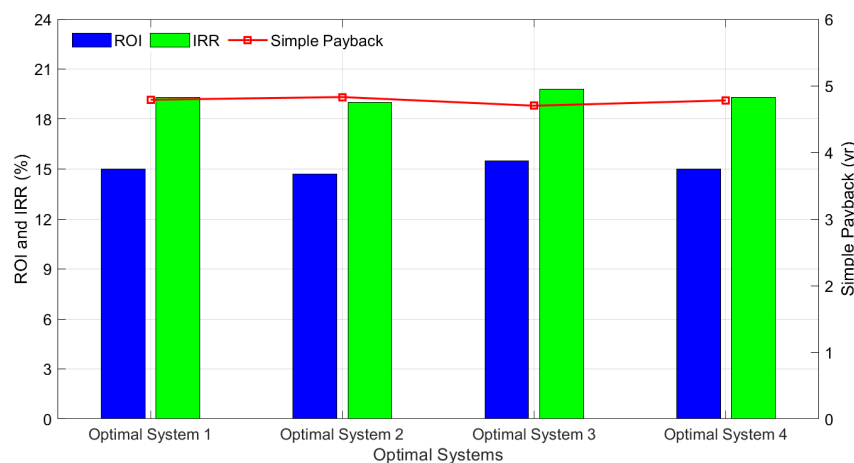


Figure 9. Comparison of return on investment (ROI), internal rate of return (IRR), and simple payback

Note: Optimal System 1, Optimal System 2, Optimal System 3, and Optimal System 4 represent the systems with no solar tracking, with horizontal-axis continuous adjustment solar tracking, with vertical-axis continuous adjustment solar tracking, and with two-axis solar tracking, respectively.

Figure 10a shows the monthly average generation of the system with no solar tracking. It can be seen that the WTs consistently share a significant part of the generation throughout the year. The seasonal contribution of the solar PV array is clear which is increasing during the spring and summer months and slightly declining in the winter months. In addition, the contribution of the DG increases during the periods of high load demand, particularly during the summer. Thus, there is a complementary relationship among the system components, where the WTs provide a relatively stable backbone, solar PV array improves the generation during the months of high solar irradiance, and DG acts as a backup resource. The power generation profile, shown in Figure 11a, reveals the hourly operational dynamics of the system components. The wind power generation varies significantly from hour to hour while the PV power peaks during the hours of daylight and goes down to zero during the night hours. The DG, on the other hand, responds to the requirements of load demand as well as the intermittent nature of renewables. Generally speaking, even though the renewable technologies supply 61.4% of load annually, the existence of a relatively large DG indicates that the peak demand and reliability requirements necessitate firm capacity that is available under all conditions. Therefore, from a technical standpoint, the system with no solar tracking achieves a substantial renewable fraction while securing operational stability and reliability.

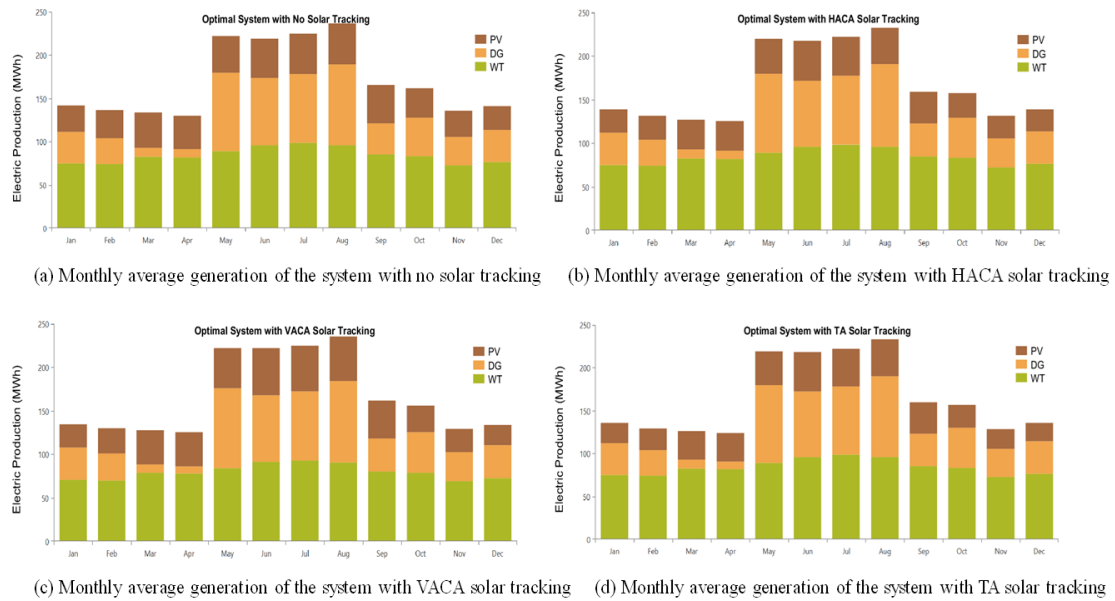


Figure 10. Monthly average generation of the system with different solar tracking strategies

Note: PV = photovoltaic; WT = wind turbine; DG = diesel generator; HACA = horizontal-axis continuous adjustment; VACA = vertical-axis continuous adjustment; TA = two-axis.

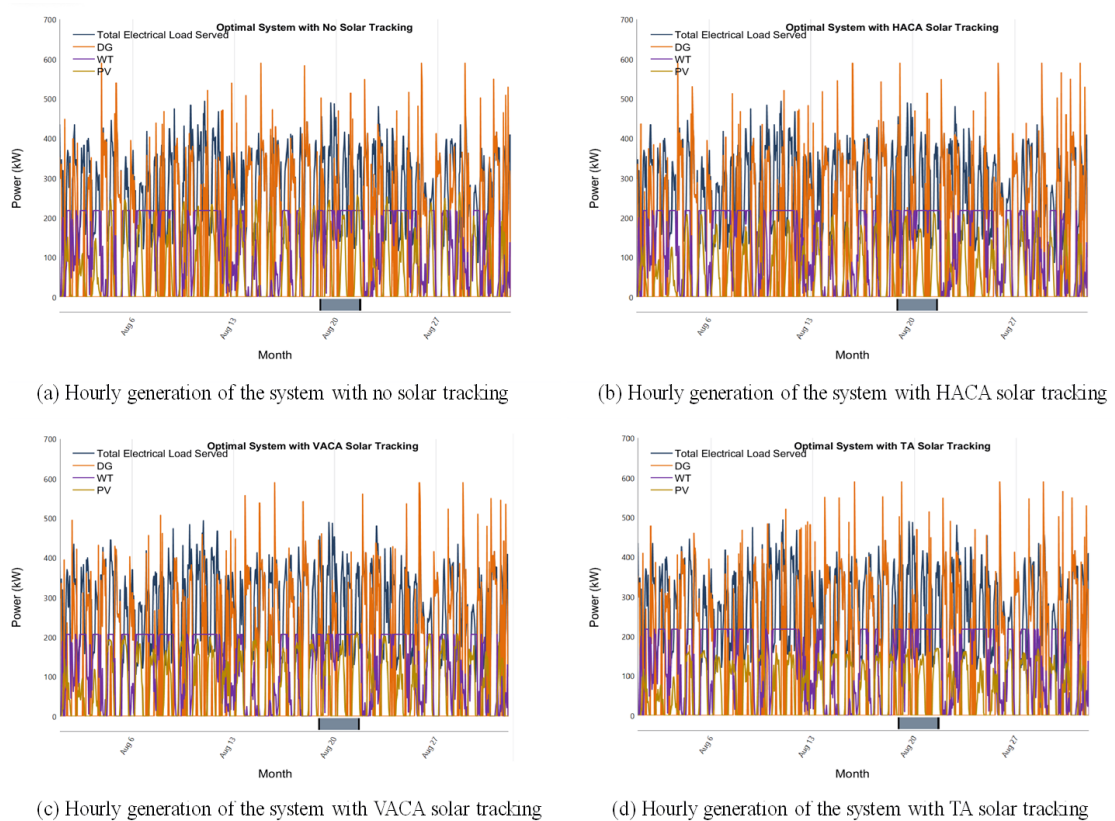


Figure 11. Hourly generation of the system with different solar tracking strategies

Note: PV = photovoltaic; WT = wind turbine; DG = diesel generator; HACA = horizontal-axis continuous adjustment; VACA = vertical-axis continuous adjustment; TA = two-axis.

Table 3 provides the capacities of the different components of the HRES with the HACA solar tracking. In this case, the system comprises a 233 kW of solar PV array, 19 WTs, a 590 kW DG, 616 battery units, and a 359 kW power converter. When compared to the no solar tracking case, the system with the HACA solar tracking strategy uses a smaller solar PV capacity. This indicates that solar irradiance capture and thereby energy production efficiency was

improved by the tracking system. As the solar PV modules follow the sun, the system demands less installed solar PV capacity to provide comparable annual energy. The number of the WTs and the size of the DG remain unchanged. This implies that the wind power still represents an essential part of the renewable generation in the system as well as the DG continues to support the system reliability. The number of batteries increased slightly from 608 to 616 units which reveals a marginal higher need for energy storage because of the changes in solar PV output introduced by the tracking system. The capacity of the power converter is almost identical between the two cases. Furthermore, the renewable fraction reduced slightly, from 61.4% in the no solar tracking case to 61.1% with the HACA solar tracking. This small reduction in the renewable fraction reveals that the tracking system primarily impacts the configuration of the system and does not lead to significant changes in renewable penetration. The monthly electricity generation of the system with the HACA solar tracking, illustrated in Figure 10b, shows that the most noticeable change is in the solar PV contribution. Despite the installed solar PV capacity is reduced by almost 18%, the monthly average energy generation from the solar PV array remains competitive with the no solar tracking case. This confirms what was stated earlier that the HACA solar tracking enhances solar irradiance capture such that the solar PV array produces comparable energy with less PV capacity. The other generation sources, i.e., the WTs and DG, continue to provide a relatively stable contribution across the year. The hourly generation profile of the system components, depicted in Figure 11b, exhibits a similar operational behavior as in the no solar tracking case. In overall, this system-level technical analysis demonstrates that the HACA solar tracking has the ability to improve solar harvesting efficiency which leads to reduced installed PV capacity without adversely impacting annual solar PV energy production.

The technical specifications of the HRES with the VACA solar tracking are provided in Table 5. The system includes a solar PV capacity of 223 kW, 18 WTs, a 590 kW DG, 608 batteries, and a 364 kW converter. Relative to the system with no solar tracking, the solar PV size decreased by 21.5%, amounting to 223 kW. This reduction in solar PV capacity is an indication of enhanced utilization of solar irradiance through the VACA solar tracking. Also, the number of WTs decreased slightly from 19 WTs in the no solar tracking case to 18 WTs with the VACA solar tracking. This reflects that introducing the VACA solar tracking led the optimization process to rebalance the share of renewable technologies in the system. The capacity of DG and the number of batteries stay unchanged. However, the capacity of the power converter slightly increased from 360 kW in the no solar tracking case to 364 kW with the VACA solar tracking. This reveals that the system processes slightly higher instantaneous power due to the variation in the solar PV output peaks. Furthermore, the renewable fraction declined slightly to 60.9% compared to 61.4% in the no solar tracking case. This indicates that the overall system balance relies on all the components collectively and not only on the solar PV efficiency. Figure 10c shows the monthly average generation of the system with the VACA solar tracking. It is apparent that the solar PV contribution stays strong, particularly during the spring and summer months, even with a smaller installed PV capacity. This reveals the capability of the VACA solar tracking in enhancing the solar capture efficiency, especially during the months of high solar irradiance. The WTs contribute consistently to the monthly electricity generation throughout the year. In addition, the DG continues to provide a relatively firm generation across the year. Therefore, the system achieves comparable electricity generation with reduced solar PV and wind capacities, which indicates an overall efficiency enhancement. The hourly generation profile, illustrated in Figure 11c, shows similar operational patterns to those observed in the no solar tracking case. This system-level analysis reveals that the VACA solar tracking can improve the solar efficiency to an extent that results in a decrease in the installed capacity of solar PV array and wind power. This efficiency improvement is evident from achieving a renewable fraction comparable to that in the no solar tracking case with smaller solar PV and wind capacities. However, it should be noted that the improvement in efficiency is not sufficient to significantly reduce the system reliance on the DG.

Table 7 provides the sizes of the different components of the HRES with the TA solar tracking. The system design in this case includes a solar PV capacity of 179 kW, 19 WTs, a 590 kW DG, 616 batteries, and a 357 kW power converter. When compared to the no solar tracking case, a 37% reduction was observed in the solar PV capacity, where the installed PV size dropped from 284 kW in the no solar tracking system to 179 kW in the TA solar tracking system. This reduction reflects a considerably higher solar capture efficiency achieved as the solar PV modules track the sun in both azimuth and elevation. Despite this large reduction in the solar PV size, the number of WTs as well as the DG capacity remain unchanged. This indicates that the wind power and DG still represent the backbone of the system through constituting a major part of the capacity structure. The system's storage requirement showed a slight increase where the number of batteries increased by 8 units compared to the no solar tracking case. The size of the power converter, on the other hand, decreased slightly by 3 kW. This reveals that the power conversion peaks do not increase even with the improved solar capture efficiency. The renewable fraction shows a slight decrease from 61.4% in the case of no solar tracking to 61% in the case of TA solar tracking. This reveals that even though the TA tracking strategy improves solar efficiency, it does not lead to a notable change in the system's renewable penetration. The monthly generation profile, illustrated in Figure 10d, shows that even with a smaller PV capacity, the solar energy contribution is competitive, particularly during the high irradiance months. The electricity generation from solar PV array remains strong from March to September, which implies that the TA

solar tracking can effectively compensate for the reduced installed PV capacity. The electricity generation from the WTs still represents the most stable renewable share across all months. The DG generation is similar to that in the no solar tracking case, which reveals that the system still necessitates substantial DG support. Figure 11d depicts the hourly power generation profile of the system components. As in the no solar tracking case, the generated power from the different components follows relatively similar patterns and successfully meets the total load in the system. In overall, the system with the TA solar tracking becomes more efficient in terms of capturing solar irradiance. This improvement leads to higher solar efficiency per installed kW, which enables the system to operate with a much smaller solar PV capacity. However, the obtained improvement in solar efficiency does not significantly alter the overall operational structure of the system.

A side by side comparison of the four system configurations reveals how the system optimal sizing changes as the solar tracking improves. In the no solar tracking case, the size of the solar PV array is the largest among all the system configurations as the fixed PV modules require more installed capacity to compensate for deficient solar irradiance capture during the day. As the HACA or VACA solar tracking strategies utilized, the solar PV array size starts to reduce, and with the TA solar tracking, the reduction becomes significant. As stated earlier, this reduction in solar PV capacity is an indication of improved solar capture efficiency because of better orientation of the PV modules with the sun. The sizes of the other generation sources within the system, i.e., the WTs and DG, remained almost constant across the four systems. From a technical perspective, this indicates that the solar tracking mechanisms do not eliminate the need for firm generation within the hybrid systems. Figure 12 depicts a comparison of the yearly electricity production of the system under various solar tracking strategies. It can be observed that all four systems produce approximately identical total yearly energy. However, the contribution of solar PV array in the system with the VACA solar tracking is relatively the highest compared to those in the other systems.

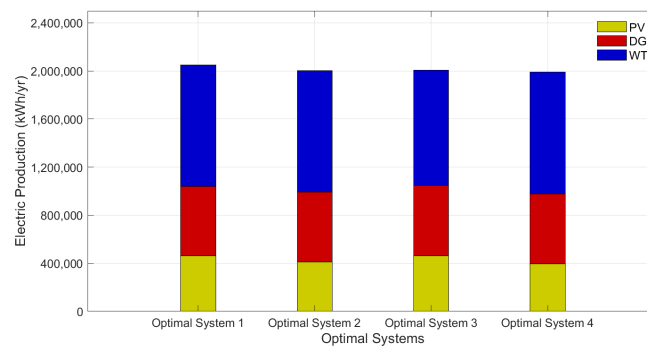


Figure 12. Comparison of yearly electricity production

Note: Optimal System 1, Optimal System 2, Optimal System 3, and Optimal System 4 represent the systems with no solar tracking, with horizontal-axis continuous adjustment solar tracking, with vertical-axis continuous adjustment solar tracking, and with two-axis solar tracking, respectively. PV = photovoltaic; WT = wind turbine; DG = diesel generator.

3.3 System-Level Environmental Analysis of Solar Trackers

It is well known that air pollutants have direct negative effects on the environment, human health, and economy. Thus, one of the critical aspects in designing a HRES is the assessment of the emissions of air pollutants from the system. In this work, an environmental analysis was conducted to determine how the different solar tracking strategies may alter the amount of pollution emitted from a HRES in the study region. Figure 13 shows a comparison of the yearly air pollutants emissions of the system under different solar tracking strategies. Since the hybrid system architecture contains a DG in every configuration, the differences observed in Figure 13 reflects how effectively each solar tracking strategy reduces the system dependency on the DG operation during the year. It shows that the values of carbon monoxide are relatively close across all the systems, sitting a little above 2500 kg/yr. The unburned hydrocarbons and particulate matter exhibit a similar pattern with the lowest level of pollution compared to the others. Additionally, in all the configurations, the sulfur dioxide is around 1000 kg/yr whereas the nitrogen oxides are around 2400 kg/yr, where the overall yearly differences are within a few kg across all the systems. The values of the carbon dioxide range approximately between 401,000 and 407,000 kg/yr. In the system with no solar tracking, the value of carbon dioxide is the lowest, while its value slightly increases with the HACA solar tracking, peaks with the VACA solar tracking, and then slightly declines with the TA solar tracking. Despite the solar tracking strategies improve solar efficiency, the dispatch optimization and renewable balancing in the system may not lead to reduce DG runtime annually. Therefore, a more advanced solar tracking does not always offer a significant environmental advantage.

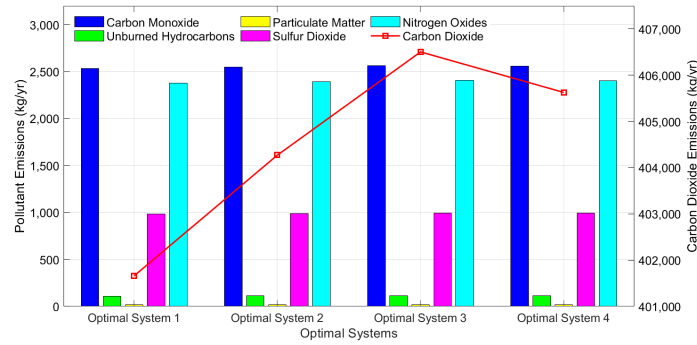
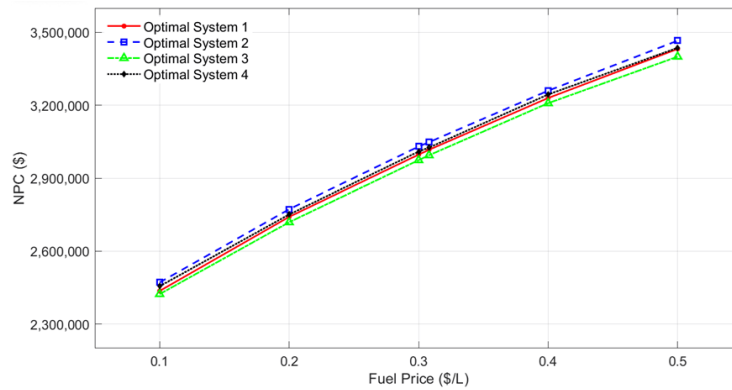


Figure 13. Comparison of yearly air pollutants emissions

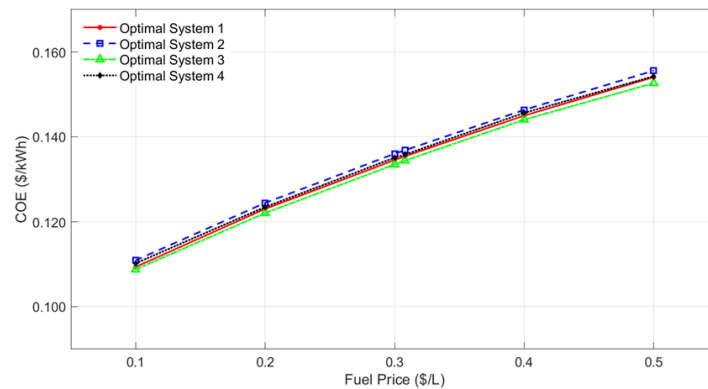
Note: Optimal System 1, Optimal System 2, Optimal System 3, and Optimal System 4 represent the systems with no solar tracking, with horizontal-axis continuous adjustment solar tracking, with vertical-axis continuous adjustment solar tracking, and with two-axis solar tracking, respectively.

3.4 Sensitivity Analysis

To further explore the economic performance of the proposed HRES with each solar tracking configuration, sensitivity analyses were conducted for both fuel price and nominal discount rate. Such analyses are important to assess not only the current economic feasibility of the solar tracking systems, but also their long-term viability and robustness under future economic conditions. Figure 14 shows the variations of the total NPC and COE with changes in fuel price from \$0.1/L to \$0.5/L.



(a) The sensitivity of the total NPC to variations in the fuel price



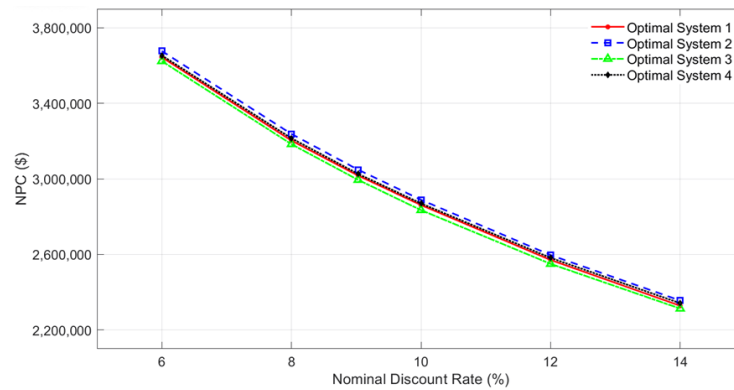
(b) The sensitivity of the COE to variations in the fuel price

Figure 14. The sensitivity of the total net present cost (NPC) and cost of energy (COE) to variations in the fuel price

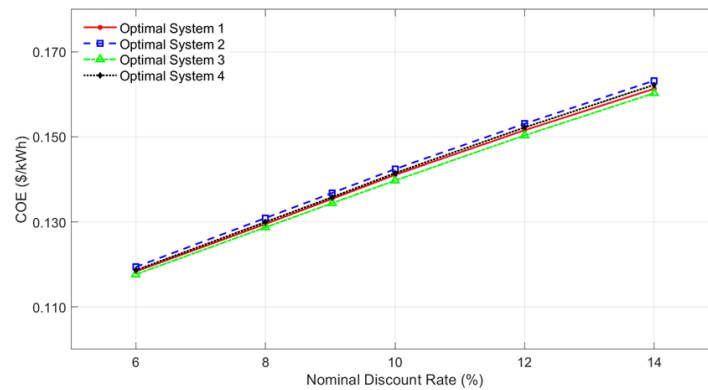
Note: Optimal System 1, Optimal System 2, Optimal System 3, and Optimal System 4 represent the systems with no solar tracking, with horizontal-axis continuous adjustment solar tracking, with vertical-axis continuous adjustment solar tracking, and with two-axis solar tracking, respectively.

It can be seen that both the total NPC and COE increase steadily across all tracking configurations as the fuel price rises. This is expected as the DG represents a crucial supporting component in the system. However, the HRES with the HACA solar tracker exhibits the highest total NPC and COE while it achieves the lowest total NPC and COE with the VACA solar tracker. In addition, the total NPC and COE of the system with no solar tracking or with the TA solar tracker remain very close to each other, with the no solar tracking case yielding relatively lower values. This is a strong indication that advanced solar tracking mechanism may not always provide economic benefits if the savings gained from reduced fuel consumption cannot offset the tracking-related costs.

Moreover, given that the nominal discount rate strongly affects the present value of future costs, the sensitivity analysis conducted on the nominal discount rate provides significant insights regarding long-term investment decisions in HRESs. Figure 15 presents the assessment of each solar tracking configuration at the system level under multiple nominal discount rates. It can be observed that there are two contradictory trends as the nominal discount rate varies from 6% to 14%. First, the total NPC across all tracking configurations decreases as the nominal discount rate increases. This is due to the fact that higher nominal discount rate values reduce the present value of future operating, replacement, and maintenance costs. Second, the COE of the HRES with all tracking configurations increases as the nominal discount rate rises. This is because at higher nominal discount rate, the future revenues of the HRES are valued less at the present terms. Most importantly, the HRES with the VACA solar tracker maintains the lowest total NPC and COE values across the entire range of nominal discount rates, followed by the no solar tracking case, then the TA tracking case, while the HACA tracking case remains the most expensive option. This analysis confirms the prior economic findings that the VACA solar tracking system represents the most attractive tracking configuration capable of achieving balance between system performance improvement and additional tracking-related costs.



(a) The sensitivity of the total NPC to variations in the nominal discount rate



(b) The sensitivity of the COE to variations in the nominal discount rate

Figure 15. The sensitivity of the total net present cost (NPC) and cost of energy (COE) to variations in the nominal discount rate

Note: Optimal System 1, Optimal System 2, Optimal System 3, and Optimal System 4 represent the systems with no solar tracking, with horizontal-axis continuous adjustment solar tracking, with vertical-axis continuous adjustment solar tracking, and with two-axis solar tracking, respectively.

3.5 Final Discussion and Evaluation of Findings

Even though all solar tracking configurations enhanced the solar capture efficiency of the solar PV array within the system, the results reveal that the system with the VACA solar tracking yielded superior overall performance. This finding indicates the capability of VACA solar tracking to attain an effective balance between energy gain and complexity of the system. In comparison with the no solar tracking case, the VACA solar tracking system improved the utilization of solar resource at the site and enabled the optimization process to reduce the required capacity of solar PV array from 284 kW to 223 kW as well as the number of WTs from 19 to 18 units while keeping energy production comparable. This resulted in an improvement in the overall economics of the HRES and led to the lowest total NPC and COE. On the other hand, although the HACA and TA solar tracking strategies exhibit improved solar harvesting compared with the no solar tracking case, their extra investment and maintenance requirements outweigh their added energy gain benefits. Therefore, according to the findings, the best solar tracking strategy within HRESs is not necessarily associated with maximizing solar irradiance capture, but is determined based on its ability to achieve the best balance between energy gains, component sizing, and lifecycle expenditure. This balance was achieved by the VACA solar tracking strategy in this work and thereby it provided superior overall performance.

4 Conclusion

In this work, an optimal design of a HRES that integrates solar PV array, WTs, DG, battery storage, and power converter to meet a residential load demand in the city of Dakuk in Iraq was studied under different solar tracking strategies. Specifically, four different configurations of the HRES were examined, including the system with no solar tracking, the system with the HACA solar tracking, the system with the VACA solar tracking, and the system with the TA solar tracking. The main goal was to explore and quantify the system-level impacts of each solar tracking strategy from economic, technical, and environmental perspectives. To that end, comprehensive analyses were performed through the assessment and comparison of all the configurations under the same load demand and resource conditions. The results showed that the HRES with the VACA solar tracking is the most cost-effective among all the configurations. It has the lowest total NPC and COE, with \$2.99 million and 0.134 \$/kWh, respectively. This indicates that the VACA solar tracking has the potential to improve solar capture to the point that the electricity produced from the solar PV array compensates for the extra capital investment and operation and maintenance cost of the tracker. In addition, the analysis demonstrated that the VACA solar tracking can accelerate the system's capital recovery. The HRES incorporating the VACA solar tracking achieved the highest ROI and IRR, at 15.5% and 19.8% respectively, and the shortest simple payback period at 4.7 years. Technically, the results showed that the optimal sizing of the system can change as the solar tracking strategy changes. The system possesses the largest size of solar PV array with no solar tracking, whereas it has the smallest solar PV array size with the TA solar tracking. As solar capturing improves, the capacity of solar PV array decreases while its contribution to electricity generation becomes more efficient per kW installed. Further, the findings showed that the sizes of the WTs and DG remain almost unchanged among the four configurations, indicating that the solar tracking strategies do not eliminate the need for firm generation within the hybrid systems. Environmentally, the results demonstrated that the pollutants emissions varied marginally among the four configurations. This indicates that advanced solar tracking systems may not always provide considerable environmental advantages. Overall, based on the findings and the conducted analyses, this work suggests that the VACA solar tracking represents the most advantageous option for use in Iraq. The methodology and the analyses conducted, as presented in this work, could be followed both locally and globally to assess the system-level viability of solar tracking strategies within HRESs.

Author Contributions

Conceptualization, A.M.S.A.; methodology, A.M.S.A.; software, A.M.S.A. and H.H.A.; validation, A.M.S.A. and H.H.A.; formal analysis, A.M.S.A.; investigation, A.M.S.A.; resources, A.M.S.A. and H.H.A.; data curation, A.M.S.A.; writing—original draft preparation, A.M.S.A.; writing—review and editing, A.M.S.A. and H.H.A.; visualization, A.M.S.A. and H.H.A.; supervision, A.M.S.A.; project administration, A.M.S.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Nomenclature

PV	photovoltaic
WT	wind turbine
DG	diesel generator
HRES	hybrid renewable energy system
PSO	particle swarm optimization
HACA	horizontal-axis continuous adjustment
VACA	vertical-axis continuous adjustment
TA	two-axis
HOMER	hybrid optimization of multiple energy resources
NPC	net present cost
COE	cost of energy
ROI	return on investment
IRR	internal rate of return

$C_{\text{ann,tot}}$	total annualized cost
i	real discount rate
n	project lifetime
CRF	capital recovery factor
E_{served}	total yearly electrical demand met
$C_{\text{i, base}}$	nominal annual cash flow of the base system
C_{i}	nominal annual cash flow of the proposed system
C_{cap}	capital cost of the proposed system
$C_{\text{cap,base}}$	capital cost of the base system
Y_{PV}	rated capacity of the PV array
f_{PV}	PV derating factor
G_{T}	instantaneous value of the solar radiation
$G_{\text{T,STC}}$	solar radiation at standard test conditions
α_{P}	temperature coefficient of power
T_{C}	instantaneous value of PV cell temperature
$T_{\text{C,STC}}$	PV cell temperature under standard test conditions
P	wind turbine power output
ρ	air density
C_{P}	wind turbine power coefficient
A	rotor swept area
V	wind speed at the wind turbine hub height